Class 29: Reciprocal Space – 3: Ewald sphere, Simple Cubic, FCC and BCC in Reciprocal Space

We have seen that diffraction occurs when, in reciprocal space,

$$\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} = \overrightarrow{H_{hkl}}$$

Let us now plot this information. Let us designate 'O' as the origin of reciprocal space, and draw the incident beam to contact the origin. Therefore, $\overrightarrow{AO}$ is the incident beam, that has the magnitude $\frac{1}{\lambda}$, as shown in the Figure 29.1 below and is therefore $\frac{\overrightarrow{S_0}}{\lambda}$. With 'A' as the center (not 'O'), let us draw a circle of radius $\frac{1}{\lambda}$ (We are drawing a circle in 2D, but a more complete picture is that of a sphere in 3D). Consider the vector $\overrightarrow{AB}$, it has the magnitude $\frac{1}{\lambda}$, which is the same as the incident beam, but has a different direction. To see if diffraction occurs in that direction, let us first designate $\overrightarrow{AB}$ as $\frac{\vec{S}}{\lambda}$. The vector $\overrightarrow{OB}$ is therefore $\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda}$, and represents the left hand side of the equation above. All points on the circle in 2D, or sphere in 3D, represent different directional possibilities of the left hand side of the equation above.

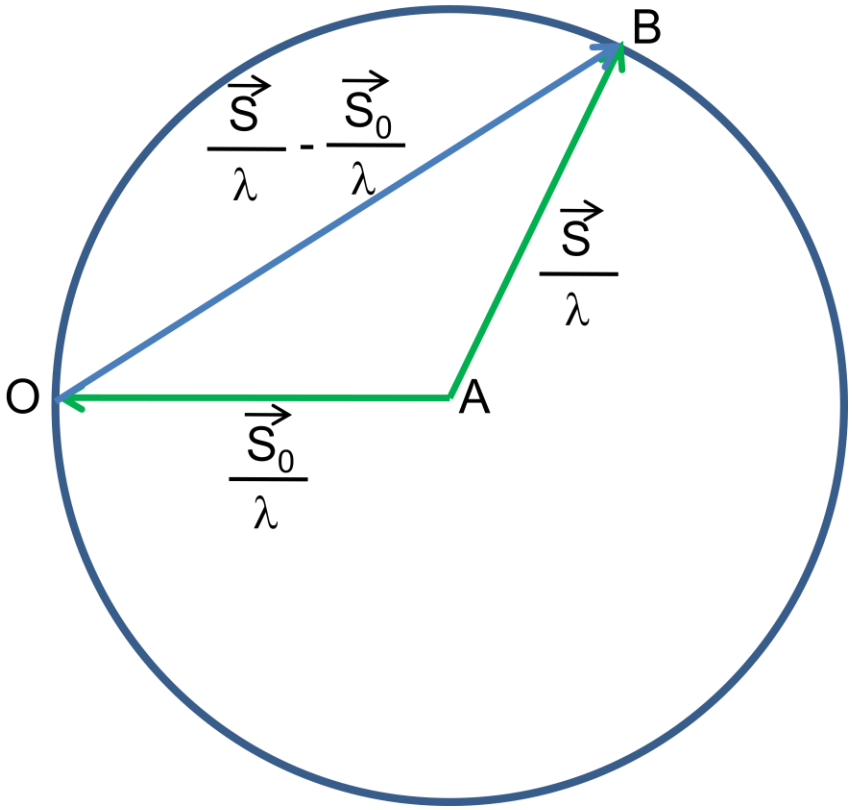


Figure 29.1: Representation of the left hand side of the equation for diffraction in reciprocal space.

The equation suggests that diffraction will occur if OB for the material interacting with the waves indicated in Figure 29.1 above.

Therefore, on this same plot we also need to draw the reciprocal lattice corresponding to the material to the same scale. Taken together, it will be possible to identify the directions at which diffraction will occur.

Starting from the origin of reciprocal space, let us plot all combinations of b_1 , i.e. the reciprocal lattice points. All the points obtained in this manner will therefore represent valid H_{hkl} vectors, or the right hand side of the equation. Figure 29.2 below therefore represents the information of the waves, as well as the periodic structure of the material in a single plot in reciprocal space.

$\overrightarrow{\quad}$ turns out to be a valid $\overrightarrow{H_{hkl}}$
$\overrightarrow{\quad}$, $\overrightarrow{b_2}$ and $\overrightarrow{b_3}$

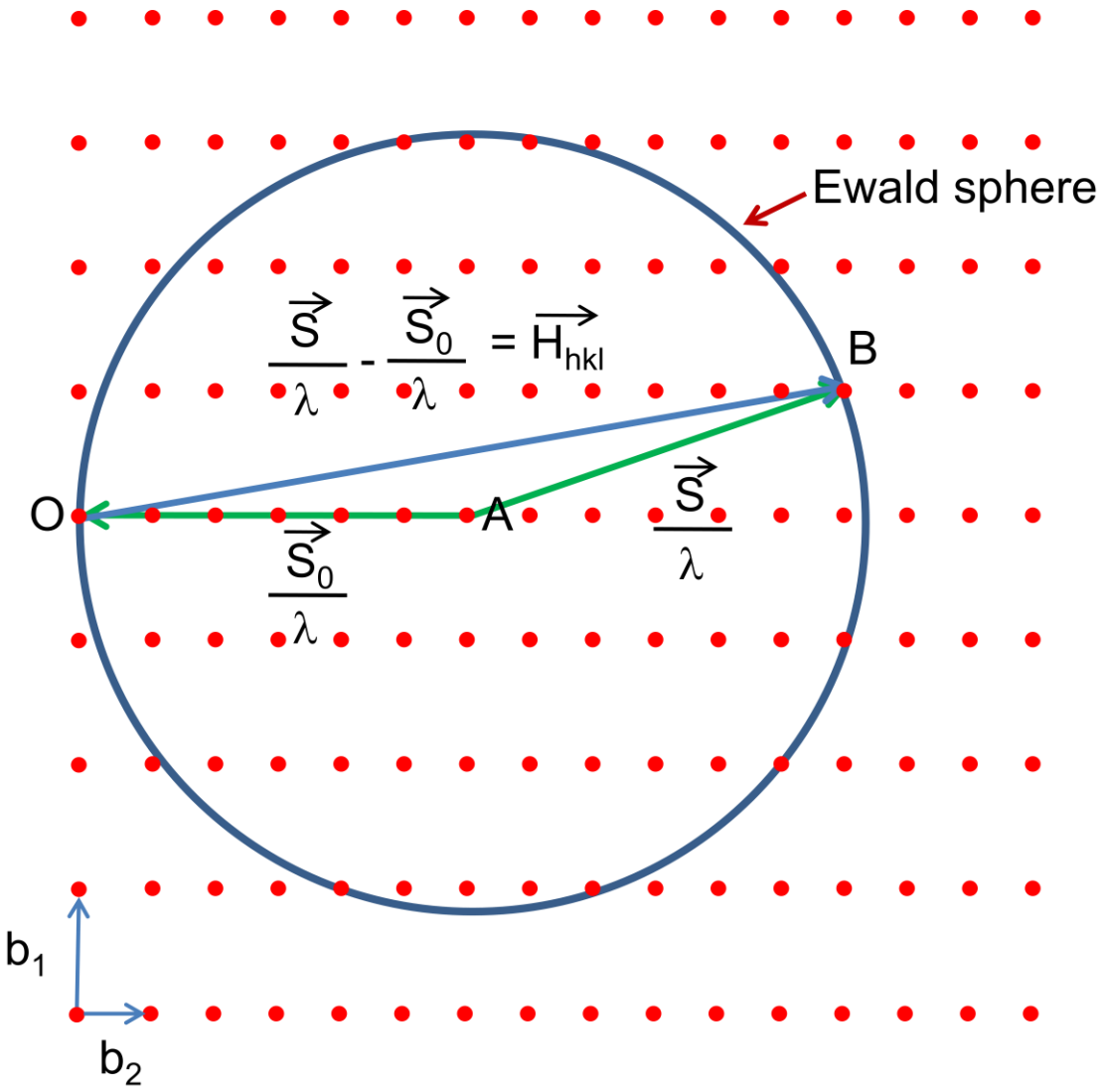


Figure 29.2: Information of the waves, as well as the periodic structure of the material in a single plot in reciprocal space. The sphere corresponding to the waves is called the Ewald sphere or sphere of reflection.

Any direction in which the sphere touches a reciprocal lattice point corresponds to a situation where the entire equation is satisfied simultaneously, i.e.

$$\frac{\vec{S}}{\lambda} - \frac{\overrightarrow{S_0}}{\lambda} = \overrightarrow{H_{hkl}}$$

And therefore the condition for diffraction is satisfied, and diffraction will occur. For all other points on Figure 29.2, either the right hand side of the equation is not satisfied (no valid reciprocal lattice point is present), or the left hand side of the equation is not satisfied

(sphere/circle is not present), or both are not satisfied. Therefore at all such points diffraction condition is not satisfied, and hence diffraction will not occur.

It is important to note that the sphere and the lattice can be selected independently – i.e. independent of each other. We can choose any wavelength λ to carry out the study – thereby we can select a sphere of our choice. We can also choose any material to study, therefore we can select the reciprocal lattice of our choice. For a given pair of λ and material, the above procedure will show us the directions at which diffraction will occur.

Let us now look at the reciprocal lattice more closely to understand how structures get represented in it and its implications.

The real lattice vectors $\overrightarrow{a_1}, \overrightarrow{a_2}$, and $\overrightarrow{a_3}$, give us the reciprocal lattice vectors $\overrightarrow{b_1}, \overrightarrow{b_2}$ and $\overrightarrow{b_3}$, based on the relationship of the kind:

$$\overrightarrow{b_1} = \frac{\overrightarrow{a_2} \times \overrightarrow{a_3}}{V}$$

A real material may have a specific structure such as simple cubic structure or FCC or BCC etc, for which there will be corresponding vectors $\overrightarrow{a_1}, \overrightarrow{a_2}$, and $\overrightarrow{a_3}$. It is of interest to see how such a material will be depicted in reciprocal space, or in other words, what will the corresponding values of $\overrightarrow{b_1}, \overrightarrow{b_2}$ and $\overrightarrow{b_3}$ be?

Let us look at some specific cases and examine the results obtained when the structure is represented in reciprocal space.

Consider a simple cubic structure. The unit vectors will have the same magnitude in all three directions.

Therefore:

$$\overrightarrow{a_1} = a\hat{x}$$

$$\overrightarrow{a_2} = a\hat{y}$$

$$\overrightarrow{a_3} = a\hat{z}$$

Where $\hat{x}, \hat{y}$, and $\hat{z}$ are unit vectors in the x, y, and z directions respectively.

The volume of the unit cell is given by:

$$V = \overrightarrow{a_1}.(\overrightarrow{a_2} \times \overrightarrow{a_3}) = a\hat{x}.(a\hat{y} \times a\hat{z}) = a\hat{x}.(a^2\hat{x}) = a^3$$

Therefore:

$$\overrightarrow{b_1} = \frac{a^2\hat{x}}{a^3} = \frac{1}{a}\hat{x}$$

$\overrightarrow{b_1}$ therefore has the magnitude $\frac{1}{a}$, and still has the direction $\hat{x}$

By symmetry we have:

$$\overrightarrow{b_2} = \frac{1}{a}\hat{y}$$

$$\overrightarrow{b_3} = \frac{1}{a}\hat{z}$$

Therefore, a simple cubic structure of side a in real space is represented by a simple cube in reciprocal space. Only the magnitude of the side has changed.

Let us now look at a FCC structure in real space. By convention the unit vectors of choice connect the origin to the three face centers. Therefore:

$$\overrightarrow{a_1} = \frac{a}{2}(\hat{y} + \hat{z})$$

$$\overrightarrow{a_2} = \frac{a}{2}(\hat{x} + \hat{z})$$

$$\overrightarrow{a_3} = \frac{a}{2}(\hat{x} + \hat{y})$$

Therefore

$$V = \overrightarrow{a_1}.(\overrightarrow{a_2} \times \overrightarrow{a_3}) = \frac{a^3}{4}$$

And

$$\overrightarrow{b_1} = \frac{\overrightarrow{a_2} \times \overrightarrow{a_3}}{V}$$

Gives us:

$$\overrightarrow{b_1} = \frac{1}{a}(-\hat{x} + \hat{y} + \hat{z})$$

Similarly, by symmetry

$$\overrightarrow{b_2} = \frac{1}{a}(\hat{x} - \hat{y} + \hat{z})$$

$$\overrightarrow{b_3} = \frac{1}{a}(\hat{x} + \hat{y} - \hat{z})$$

A plot of these vectors will result in a BCC structure, where these are vectors that connect the origin to the nearest three body centers. It is important to note that the material has not changed its crystal structure, it just so happens that the representation of a FCC structure in reciprocal space shows the attributes of a BCC structure.

Finally, let us consider a BCC structure in real space. By convention the unit vectors of choice connect the origin to the nearest body centers.

$$\overrightarrow{a_1} = \frac{a}{2}(-\hat{x} + \hat{y} + \hat{z})$$

$$\overrightarrow{a_2} = \frac{a}{2}(\hat{x} - \hat{y} + \hat{z})$$

$$\overrightarrow{a_3} = \frac{a}{2}(\hat{x} + \hat{y} - \hat{z})$$

This gives us the following:

$$V = \overrightarrow{a_1}.(\overrightarrow{a_2} \times \overrightarrow{a_3}) = \frac{a^3}{2}$$

And

$$\overrightarrow{b_1} = \frac{\overrightarrow{a_2} \times \overrightarrow{a_3}}{V} = \frac{1}{a}(\hat{y} + \hat{z})$$

Similarly, by symmetry

$$\overrightarrow{b_2} = \frac{1}{a}(\hat{x} + \hat{z})$$

$$\overrightarrow{b_3} = \frac{1}{a}(\hat{x} + \hat{y})$$

These represent vectors from the origin to three nearby face centers. Therefore a body centered structure in real space is represented by a face centered structure in reciprocal space.

Therefore, we have seen examples of how a material with a given structure in real space, should be represented in reciprocal space. If we combine this information with the Ewald sphere corresponding to the waves being used, we can determine if diffraction will occur.

We will use this information to understand how waves corresponding to electrons in a material, interact with the crystal structure of the same material.